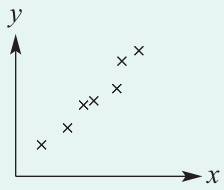
- Types of correlation:
  - The correlation is positive if the y variable generally increases as the x variable increases.
  - The correlation is negative if the y variable generally decreases as the x variable increases.

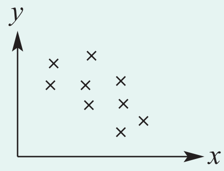
  Examples:

Strong positive correlation



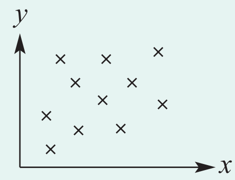
As x increases, y clearly increases.

Weak negative correlation



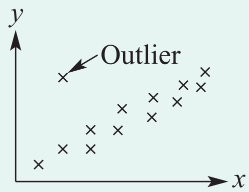
As x increases, y generally decreases.

No correlation



As x increases, there is no particular effect on y.

- An outlier can clearly be identified as a data point that is isolated from the rest of the data.



Exercise 5G

Understanding

1 Decide whether it is likely or unlikely that there will be a strong relationship between these pairs of variables.
   a Height of door and width of door
   b Weight of car and fuel consumption
   c Temperature and length of phone calls
   d Colour of flower and strength of perfume
   e Amount of rain and size of vegetables in the vegetable garden

2 For each of the following sets of bivariate data with variables x and y:
   i draw a scatter plot by hand
   ii decide whether y generally increases or decreases as x increases

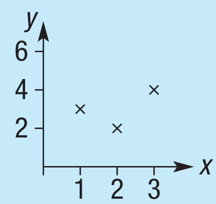
a

x	1	2	3	4	5	6	7	8	9	10
y	3	2	4	4	5	8	7	9	11	12

b

x	0.1	0.3	0.5	0.9	1.0	1.1	1.2	1.6	1.8	2.0	2.5
y	10	8	8	6	7	7	7	6	4	3	1

On a scatter plot, mark each point of the plot with a ×.



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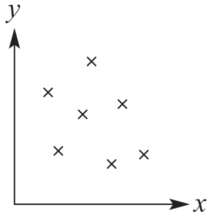
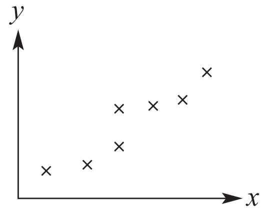
3 For these scatterplots, choose two words from those listed below to best describe the correlation between the two variables.

strong *weak* *positive* *negative*

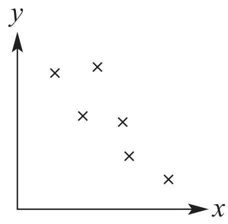
a



b



c



Fluency

Example 15 Constructing and interpreting scatter plots

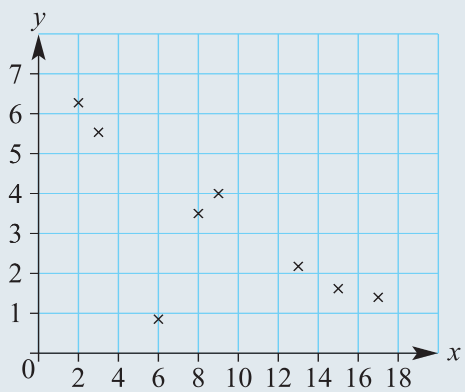
Consider this simple bivariate data set.

x	13	9	2	17	3	6	8	15
y	2.1	4.0	6.2	1.3	5.5	0.9	3.5	1.6

a Draw a scatter plot for the data.
b Describe the correlation between x and y as positive or negative.
c Describe the correlation between x and y as strong or weak.
d Identify any outliers.

Solution / Explanation

a



Draw an appropriate scale on each axis by looking at the data:
- x is up to 17
- y is up to 6.2

The scale must be spread evenly on each axis.
Plot each point using a × symbol on graph paper.

b Negative correlation — Looking at the scatterplot, as x increases y decreases.

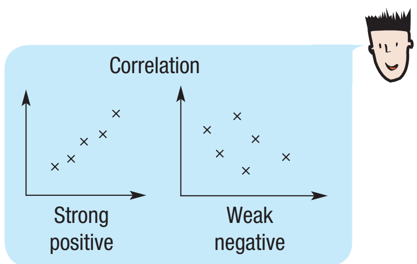
c Strong correlation — The downwards trend in the data is clearly defined.

d The outlier is (6, 0.9). — This point defies the trend.

4 Consider this simple bivariate data set.

x	1	2	3	4	5	6	7	8
y	1.0	1.1	1.3	1.3	1.4	1.6	1.8	1.0

a Draw a scatter plot for the data.
b Describe the correlation between x and y as positive or negative.
c Describe the correlation between x and y as strong or weak.
d Identify any outliers.



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5 Consider this simple bivariate data set.

x	14	8	7	10	11	15	6	9	10
y	4	2.5	2.5	1.5	1.5	0.5	3	2	2

a Draw a scatter plot for the data.
b Describe the correlation between x and y as positive or negative.
c Describe the correlation between x and y as strong or weak.
d Identify any outliers.

6 By completing scatter plots for each of the following data sets, describe the correlation between x and y as 'positive', 'negative' or 'none'.

a

x	1.1	1.8	1.2	1.3	1.7	1.9	1.6	1.6	1.4	1.0	1.5
y	22	12	19	15	10	9	14	13	16	23	16

b

x	4	3	1	7	8	10	6	9	5	5
y	115	105	105	135	145	145	125	140	120	130

c

x	28	32	16	19	21	24	27	25	30	18
y	13	25	22	21	16	9	19	25	15	12

Problem-solving and Reasoning

7 A tomato grower experiments with a new organic fertiliser and sets up five separate garden beds: A, B, C, D and E. The grower applies different amounts of fertiliser to each bed and records the diameter of each tomato picked.

The average diameter of a tomato from each garden bed and the corresponding amount of fertiliser are recorded below.

Bed	A	B	C	D	E
Fertiliser (grams per week)	20	25	30	35	40
Average diameter (cm)	6.8	7.4	7.6	6.2	8.5

a Draw a scatter plot for the data with 'Diameter' on the vertical axis and 'Fertiliser' on the horizontal axis. Label the points A, B, C, D and E.
b Which garden bed appears to go against the trend?
c According to the given results, would you be confident in saying that the amount of fertiliser fed to tomato plants does affect the size of the tomato produced?

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8 For common motor vehicles, consider the two variables *Engine size* (cylinder volume) and *Fuel economy* (number of kilometres travelled for every litre of petrol).

a Do you expect there to be some relationship between these two variables?

b As the engine size increases, would you expect the fuel economy to increase or decrease?

c The following data was collected for 10 vehicles.

Car	A	B	C	D	E	F	G	H	I	J
Engine size	1.1	1.2	1.2	1.5	1.5	1.8	2.4	3.3	4.2	5.0
Fuel economy	21	18	19	18	17	16	15	20	14	11

i Does the data generally support your answers to parts **a** and **b** above?

ii Which car gives a fuel economy reading that does not support the general trend?

9 On 14 consecutive days a local council measures the volume of sound heard from a freeway at various points in a local suburb. The volume (V) of sound is recorded against the distance (d m) between the freeway and the point in the suburb.

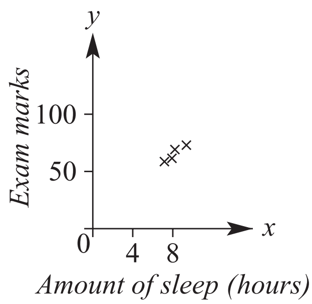
Distance (d)	200	350	500	150	1000	850	200	450	750	250	300	1500	700	1250
Volume (V)	4.3	3.7	2.9	4.5	2.1	2.3	4.4	3.3	2.8	4.1	3.6	1.7	3.0	2.2

a Draw a scatter plot of V against d, plotting V on the vertical axis and d on the horizontal axis.

b Describe the correlation between d and V as positive, negative or none.

c Generally as d increases, does V increase or decrease?

10 A person presents you with this scatter plot and suggests to you that there is a strong correlation between the amount of sleep and exam marks. What do you suggest is the problem with the person's graph and conclusions?



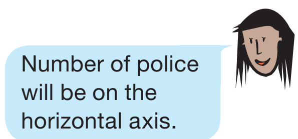
Crime rates and police

11 A government department is interested in convincing the electorate that a large number of police on patrol leads to lower crime rates. Two separate surveys are completed over a one-week period and the results are listed in this table.

	Area	A	B	C	D	E	F	G
Survey 1	Number of police	15	21	8	14	19	31	17
	Incidence of crime	28	16	36	24	24	19	21
Survey 2	Number of police	12	18	9	12	14	26	21
	Incidence of crime	26	25	20	24	22	23	19

a By using scatter plots, determine whether or not there is a relationship between the number of police on patrol and the incidence of crime, using the data in:

i survey 1 **ii** survey 2



b Which survey results do you think the government will use to make its point? Why?

Using technology 5.7: Using calculators to draw scatter plots

This activity is available on the companion website as a printable PDF.

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